

Car-following Model: Coding (Python)

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Where is the code

Popular repositories

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construct_fundamental_diagram

Public

An automated tool that can rationally construct the fundamental diagram of traffic flow from high-fidelity vehicle trajectories

Python ☆ 7 🍴 2

interactive_traffic_simulation

Public

In this (ring-road multi-lane) traffic simulation, one can take the wheel of a car for changing lanes, acceleration and deceleration in real time, also creating congestion. One can observe the traf...

MATLAB ☆ 6

traffic_flow_theory_slides

Public

Those are my (handout mode) slides in teaching traffic flow theory, which I spent a lot of time and effort

☆ 2

<https://github.com/gottrafficgo>

LLMTraveler

Public

Forked from [georgewanglz2019/LLMTraveler](#)

idm_and_traffic_wave

Public

Intelligent Driver Model + Straight Road (Open Boundary) + Stop-and-Go Waves

Python

v3

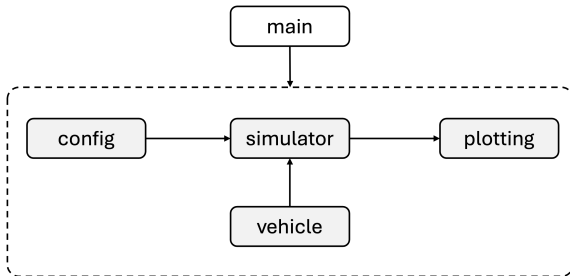
idm_and_constraints

Public

I found the additional constraints in coding largely influence the results of intelligent driver model (IDM)-based traffic flow simulation. I report some of them here.

Python

Framework



Communicate with ChatGPT

Feed *simulator._generate_vehicles()* to ChatGPT and ask

- What is the logic of this code?

Feed *all code* to ChatGPT and ask:

- Generate code to plot the position of the vehicle with $ID = 200$ over time. The x-axis should represent time, and the y-axis should represent position.
- Generate code to plot the speeds of vehicles as they pass a cross-section at $x = 1000$. The x-axis should represent time, and the y-axis should represent speed. To determine each vehicle's speed at the cross-section, loop through all vehicles and then through each vehicle's trajectory to find consecutive points where the first point is before the cross-section and the next point is after it.

Thank you!